AI PRODUCT MANAGEMENT 2 · DETAILED COURSE

Artificial Intelligence and Machine Learning Foundations

How AI and machine learning work, the main problem types and how models are trained and judged.

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CHAPTER 11

Artificial Intelligence and Machine Learning Foundations

How AI and machine learning work, the main problem types and how models are trained and judged.

Introduction to Artificial Intelligence

Systems that perform tasks associated with human intelligence, such as perception, language and decision making.

AI refers to systems that perform tasks associated with human intelligence, such as perceiving images, understanding language and making decisions. Most AI in products today is machine learning, which learns patterns from data.

History and Evolution of AI

Rules and symbolic AI, expert systems, statistical ML, deep learning from 2012, transformers from 2017, and generative AI.

The field moved from rule-based and symbolic systems to expert systems, then statistical machine learning, deep learning from around 2012, transformer models from 2017 and today's generative models. Each shift changed what products could do.

Traditional AI vs Generative AI

Traditional models classify or predict from data; generative models create new text, images, audio or code.

Traditional models classify or predict from data, such as flagging fraud. Generative models create new text, images, audio or code. Traditional AI outputs a label or number; generative AI outputs content that needs review.

Narrow AI vs General AI

Narrow AI does specific tasks; general AI would match human breadth, and does not yet exist.

Narrow AI performs specific tasks well, such as translation or recommendation. General AI would match human breadth across tasks. Only narrow AI exists today, so products should be designed around specific tasks.

AI Applications Across Industries

Fraud detection, recommendations, diagnostics, document processing, support automation, forecasting and more.

Examples include fraud detection in banking, recommendations in e-commerce, diagnostics in healthcare, document processing in insurance, support automation and demand forecasting in retail.

Machine Learning Fundamentals

Algorithms that learn patterns from data to predict or decide without explicit rules.

Machine learning algorithms learn patterns from data to predict or decide without explicit rules for every case. Quality depends mostly on data quality and on defining the target correctly.

Supervised Learning

Learning from labelled examples to predict labels for new data.

Supervised learning trains on labelled examples to predict labels for new data. Example: historical loan applications labelled repaid or defaulted, used to predict risk for a new applicant.

Unsupervised Learning

Finding structure in unlabelled data, such as clusters or anomalies.

Unsupervised learning finds structure in unlabelled data, such as clusters of similar customers or unusual transactions. Results need interpretation, because no correct answer is given.

Semi-Supervised Learning

Using a small labelled set plus a large unlabelled set.

It uses a small labelled dataset together with a large unlabelled one. This helps when labelling is expensive, such as medical images labelled only by specialists.

Reinforcement Learning

An agent learns by trial and error to maximise cumulative reward.

An agent learns by trial and error to maximise cumulative reward. It suits sequential decisions such as pricing or robot control, but needs a safe simulation or careful rollout.

Classification

Predicting a category, such as spam or not spam.

Classification predicts a category, such as spam or not spam, or churn or stay. Its output is usually a probability, which the product converts into a decision using a threshold.

Regression

Predicting a continuous number, such as price or demand.

Regression predicts a continuous number, such as delivery time or demand next week. Errors are measured as average distance from the true value.

Clustering

Grouping similar items without labels, used for segmentation.

Clustering groups similar items without labels. It is used for customer segmentation and for finding natural groups in support tickets. Clusters must be named and checked by people.

Recommendation Systems

Suggesting items using collaborative filtering, content features or hybrid models.

Recommenders suggest items using collaborative filtering (what similar users chose), content features, or hybrid models. They need careful evaluation for diversity and for avoiding narrow feedback loops.

Predictive Analytics

Using historical data to forecast outcomes such as churn or demand.

Predictive analytics uses historical data to forecast outcomes such as churn, demand or default. A forecast is only useful if someone acts on it, so define the intervention before building the model.

Feature Engineering

Creating informative inputs from raw data to help models learn.

Feature engineering creates informative inputs from raw data, such as days since last purchase. Good features often matter more than a more complex algorithm.

Training, Validation and Testing

Train to fit, validate to tune, and test once on unseen data for an honest estimate.

The model is trained on one set, tuned on a validation set and evaluated once on an unseen test set. Keeping the test set untouched gives an honest estimate of performance.

Model Training and Inference

Training learns parameters from data; inference uses the trained model to make predictions.

Training learns model parameters from data. Inference uses the trained model to make predictions on new inputs. Inference cost and latency matter in production even when training is done once.

Overfitting and Underfitting

Overfitting memorises training noise and fails on new data; underfitting is too simple to capture the pattern.

Overfitting memorises training noise and performs poorly on new data. Underfitting is too simple to capture the pattern. Compare training and validation performance to detect either.

Bias-Variance Trade-Off

Simple models have high bias; complex models have high variance; aim for the balance with best generalisation.

Simple models have high bias and miss patterns. Complex models have high variance and react to noise. The aim is the level of complexity that generalises best to new data.

Model Performance Metrics

Accuracy, precision, recall, F1, AUC for classification; MAE and RMSE for regression, picked to match business cost of errors.

Classification uses accuracy, precision, recall, F1 and AUC. Regression uses mean absolute error and root mean squared error. Pick the metric that matches the business cost of each kind of error.

Precision, Recall and F1 Score

Precision is correctness of positive predictions; recall is coverage of actual positives; F1 balances both.

Precision is the share of positive predictions that are correct. Recall is the share of actual positives found. F1 combines both. In fraud detection, recall often matters more; in automatic account blocking, precision matters more.

Confusion Matrix

A table of true/false positives and negatives that shows the types of errors.

A confusion matrix tabulates true positives, false positives, true negatives and false negatives. It shows what kinds of mistakes a model makes, which a single accuracy number hides.

ROC-AUC

Area under the curve of true-positive vs false-positive rate across thresholds; 0.5 is chance, 1.0 is perfect.

The ROC curve plots true-positive rate against false-positive rate across thresholds. AUC is the area under it: 0.5 means chance and 1.0 means perfect ranking.

Deep Learning Fundamentals

Multi-layer neural networks that learn features automatically from large data.

Deep learning uses multi-layer neural networks that learn features directly from large data, such as images or text. It needs more data and compute than classical methods.

Neural Networks

Layers of connected units with weights, trained by backpropagation and gradient descent.

Neural networks connect layers of units with weights. Training adjusts the weights with backpropagation and gradient descent to reduce error.

CNN, RNN and LSTM

CNNs process images, RNNs process sequences, LSTMs handle longer dependencies in sequences.

CNNs process images using local patterns. RNNs process sequences step by step. LSTMs handle longer dependencies in sequences. Transformers have largely replaced RNNs for language tasks.

Transformers

An architecture based on attention that processes sequences in parallel and underlies modern language models.

Transformers process sequences in parallel using attention. They underpin modern language and multimodal models, and they scale well with data and compute.

Attention Mechanisms

A way for a model to weigh which parts of the input matter for each output.

Attention lets the model weigh which parts of the input matter for each output. It is the reason models can relate a word to a distant word in the same document.

Natural Language Processing

Techniques for understanding and generating human language, from tokenisation to LLMs.

NLP covers understanding and generating human language, from tokenisation and classification to question answering. Large language models now handle most NLP tasks in products.

Computer Vision Fundamentals

Techniques for classifying, detecting and segmenting content in images and video.

Computer vision classifies, detects and segments content in images and video. Applications include document scanning, quality inspection and product tagging.

AI Model Lifecycle

Problem framing, data, training, evaluation, deployment, monitoring and retraining.

The lifecycle covers problem framing, data collection, training, evaluation, deployment, monitoring and retraining. The PM owns framing, the success definition and the monitoring plan.

Summary of terms

Term	Meaning
Introduction to Artificial Intelligence	Systems that perform tasks associated with human intelligence, such as perception, language and decision making.
History and Evolution of AI	Rules and symbolic AI, expert systems, statistical ML, deep learning from 2012, transformers from 2017, and generative AI.
Traditional AI vs Generative AI	Traditional models classify or predict from data; generative models create new text, images, audio or code.
Narrow AI vs General AI	Narrow AI does specific tasks; general AI would match human breadth, and does not yet exist.
AI Applications Across Industries	Fraud detection, recommendations, diagnostics, document processing, support automation, forecasting and more.
Machine Learning Fundamentals	Algorithms that learn patterns from data to predict or decide without explicit rules.
Supervised Learning	Learning from labelled examples to predict labels for new data.

Unsupervised Learning	Finding structure in unlabelled data, such as clusters or anomalies.
Semi-Supervised Learning	Using a small labelled set plus a large unlabelled set.
Reinforcement Learning	An agent learns by trial and error to maximise cumulative reward.
Classification	Predicting a category, such as spam or not spam.
Regression	Predicting a continuous number, such as price or demand.
Clustering	Grouping similar items without labels, used for segmentation.
Recommendation Systems	Suggesting items using collaborative filtering, content features or hybrid models.
Predictive Analytics	Using historical data to forecast outcomes such as churn or demand.
Feature Engineering	Creating informative inputs from raw data to help models learn.
Training, Validation and Testing	Train to fit, validate to tune, and test once on unseen data for an honest estimate.
Model Training and Inference	Training learns parameters from data; inference uses the trained model to make predictions.
Overfitting and Underfitting	Overfitting memorises training noise and fails on new data; underfitting is too simple to capture the pattern.
Bias-Variance Trade-Off	Simple models have high bias; complex models have high variance; aim for the balance with best generalisation.
Model Performance Metrics	Accuracy, precision, recall, F1, AUC for classification; MAE and RMSE for regression, picked to match business cost of errors.
Precision, Recall and F1 Score	Precision is correctness of positive predictions; recall is coverage of actual positives; F1 balances both.
Confusion Matrix	A table of true/false positives and negatives that shows the types of errors.
ROC-AUC	Area under the curve of true-positive vs false-positive rate across thresholds; 0.5 is chance, 1.0 is perfect.
Deep Learning Fundamentals	Multi-layer neural networks that learn features automatically from large data.
Neural Networks	Layers of connected units with weights, trained by backpropagation and gradient descent.
CNN, RNN and LSTM	CNNs process images, RNNs process sequences, LSTMs handle longer dependencies in sequences.
Transformers	An architecture based on attention that processes sequences in parallel and underlies modern language models.
Attention Mechanisms	A way for a model to weigh which parts of the input matter for each output.
Natural Language Processing	Techniques for understanding and generating human language, from tokenisation to LLMs.
Computer Vision Fundamentals	Techniques for classifying, detecting and segmenting content in images and video.
AI Model Lifecycle	Problem framing, data, training, evaluation, deployment, monitoring and retraining.

Practice exercise: Train a simple classifier on a public dataset and report precision, recall and F1 at two thresholds. Say which one the business should pick.

Self-check

1. Explain Introduction to Artificial Intelligence in two sentences to a non-technical stakeholder, then say what you would measure for it.
2. Explain Training, Validation and Testing in two sentences to a non-technical stakeholder, then say what you would measure for it.
3. Explain AI Model Lifecycle in two sentences to a non-technical stakeholder, then say what you would measure for it.

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